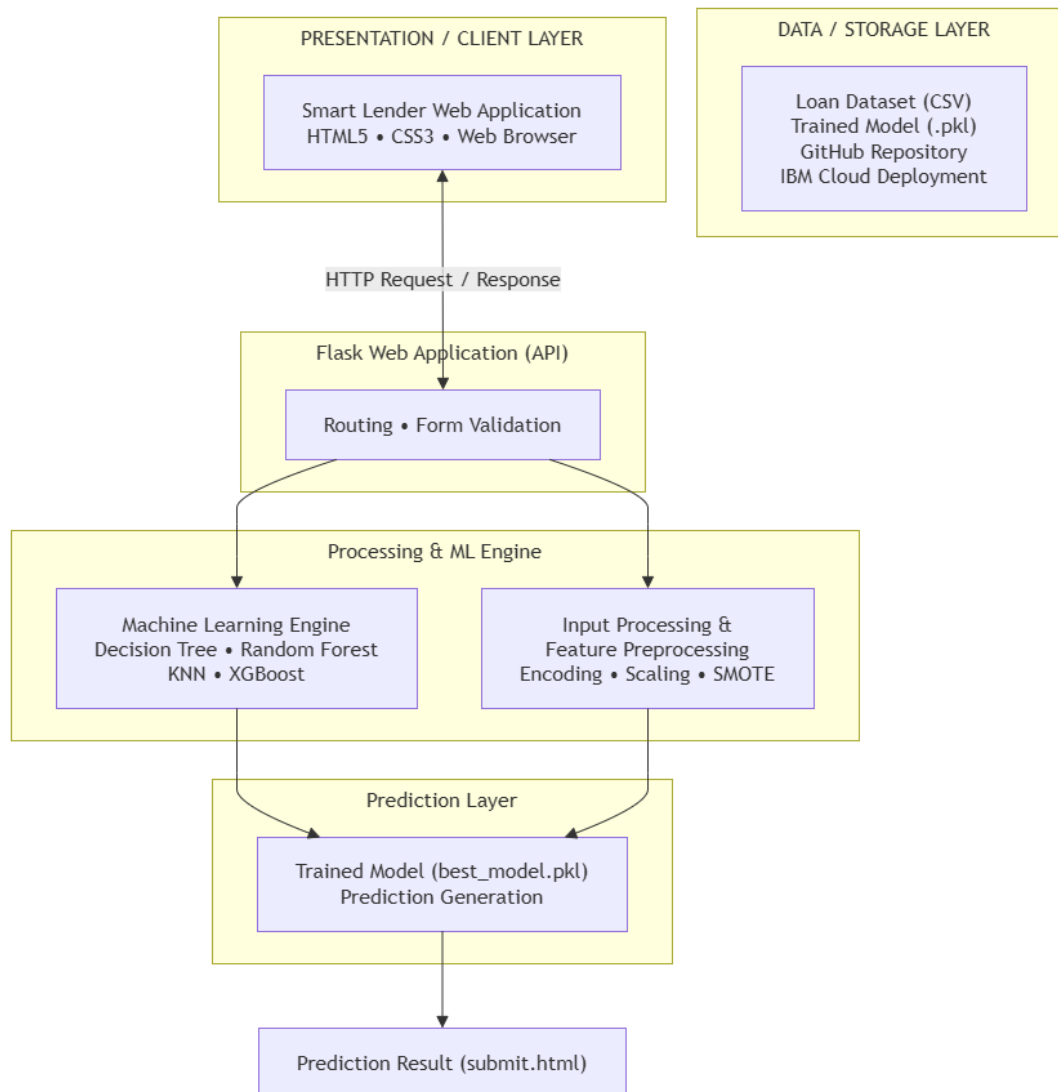


Solution Architecture

Date	15 March 2026
Team ID	XXXXXX
Project Name	Smart Lender
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	<i>Provides the graphical user interface through which users interact with the Smart Lender application. Users enter loan applicant details and view prediction results using web pages.</i>	<i>HTML5, CSS3, Web Browser</i>
API Gateway / Application Layer	<i>The Flask application receives HTTP requests, validates user input, manages routing between web pages, invokes the machine learning model, and returns prediction results to the frontend.</i>	<i>Python 3.x, Flask</i>
Core Logic Service (Machine Learning)	<i>Performs data preprocessing, feature encoding, scaling, and executes the trained machine learning model to predict loan approval. Multiple algorithms are evaluated during development, and the best-performing model is used for deployment.</i>	<i>Pandas, NumPy, Scikit-learn, XGBoost, imbalanced-learn (SMOTE), Pickle</i>
External API Integrations	<i>The current implementation operates independently without mandatory external APIs. The architecture is designed to support future integration with banking systems, credit score verification services, SMS, email notifications, or payment gateways.</i>	<i>IBM Cloud Services (Future Integration), REST APIs</i>
Data / Storage Layer	<i>Stores the loan prediction dataset used for training, the serialized machine learning model, project source code, and deployment resources.</i>	<i>CSV Dataset, Pickle (.pkl), GitHub, IBM Cloud</i>